Viscon®

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RTU



Viscon Dua Remote Terminal Unit is a revolutionary approach product in Remote Terminal Unit (RTU) design for real-time applications. Designed using the latest technological advancements, it is the best solution for larger applications such as power distribution supervisory control and data acquisition system (SCADA).

Like its predecessor, Viscon Dua RTU is designed with the concept of modularity, robust and fault tolerant. These ensure no miss-operations or failures which will affect the plant. At the same time, the RTU is able to absorb or block any plant failure such as over-voltages or currents from invading the system.

A typical unit comprises of a Central Processing Unit (CPU) card and up to three I/O cards. There are three types of I/O card which are Digital Input (DI), Digital Output (DO) and Analog Input (AI). This allows Viscon Dua RTU to be easily configured and interchanged, such as from an RTU with dedicated I/O type to mixed I/O type. Thus, ensuring that an optimum configuration is provided to best suit users demands and requirements at any given situation.

For safety, all of the RTU's built-in-plant interfaces are optically and relay isolated, providing a very robust and well protected unit, thus, enhancing the reliability of the RTU throughout its lifetime.

Product Highlights

- Small or large systems
- Modular and distributed
- More than 1000 points
- Various plant signal interface
- Various communication interfaces
- Wall, pole or rack-mount
- Industry standard compliant

General Features

- Multi input/output (analog/digital)
- Built-in protection circuits
- Point-to-point, multi-drop configuration
- Time synchronisation
- Wide power operating range
- User friendly configuration and diagnostic tools

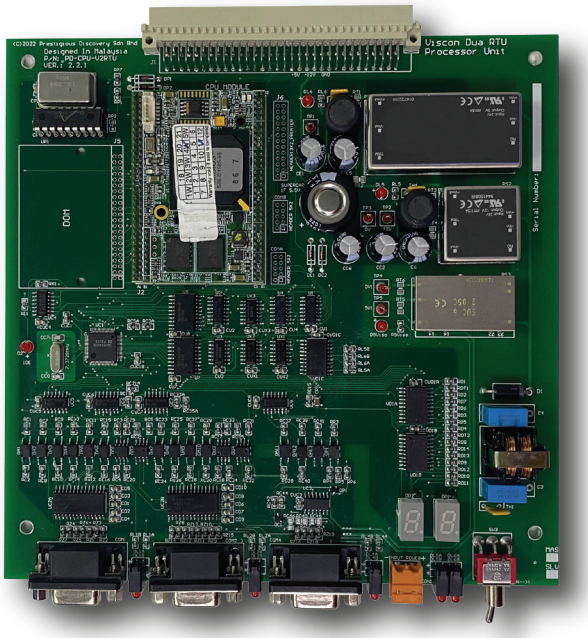
Product Overview

In general, the RTU is capable of:-

- Operating in single-port or multi-port (Master/slave) configurations for extended IOs. Each RTU supports up to 16 ports in multi-port (Master/slave) configuration.
- Monitoring of 16-channel Analog Input per card, 32 points of Digital Input per card and controlling 16 points of digital output per card.
- Communicating with SCADA Master through its RS232 port with programmable data speed. Message housekeeping and function processing are performed by the master CPU card.
- Communicating with Viscon Dua Programming and Diagnostic Utility (PDU) through dedicated diagnostic port.

Viscon Dua RTU module consists of the CPU card and up to three input/output cards either Digital Input (DI) card, Digital Output (DO) card Analog Input (AI) card and /or Analog Output Card. CPU card and I/O card are interconnected together by backplane system (motherboard). The module can have dedicated I/O card configuration i.e. DI only, DO only, AI only and AO only or the combination of the four types i.e. DI, DO, AI and AO in one module.





CPU Card

The CPU card consists of a microcomputer or micro-controller with peripherals, memories, communication interfaces, real-time clock, watchdog timer, and decoders.

The essential tasks of the CPU card are:

- Managing and controlling of the I/O cards via peripheral bus.
- Reading the process events from digital input cards.
- Reading analog input from analog input cards.
- Sending control commands to the digital output cards.
- Handling communication to the master system.
- Self test the condition of I/O cards and CPU card.

CPU Card Technical Brief

Processor And Memory

- Processor: Vortex 86SX, 300MHz
- System memory: 128MB DDR2 Onboard
- Flash memory: Onboard 2MB SPI
- Data Bus size: 16Bit x-ISA Interface

Standard Protocol

- IEC 60870-5-101

Serial Interface

- Master port
  - COM1: RS232 with Tx and RX LEDs
  - Baudrate: 9600 by default. Configurable from 300 to 115200 bits/sec
- Diagnostic port
  - COM2: RS232 with Tx and RX LEDs
  - Baudrate: 115200 bits/sec
- Link port
  - RS422 with Tx and Rx LEDs

Power Supply

- 24DC nominal
- Power LEDs

Mechanical Layout

- Printed circuit board: 186 mm x 190 mm
- Subrack: 2U x 19”

Connector types

- Master port: DB9 Female
- Diagnostic port: DB9 Female
- Link port: DB9 Female

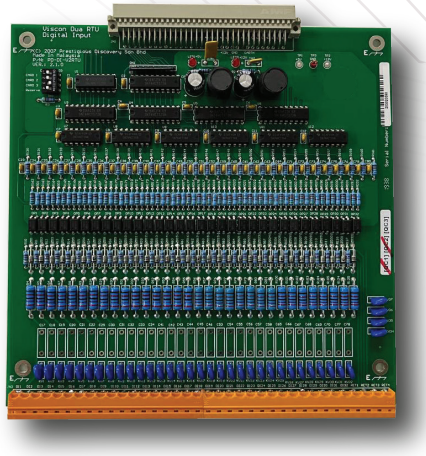
Environment conditions

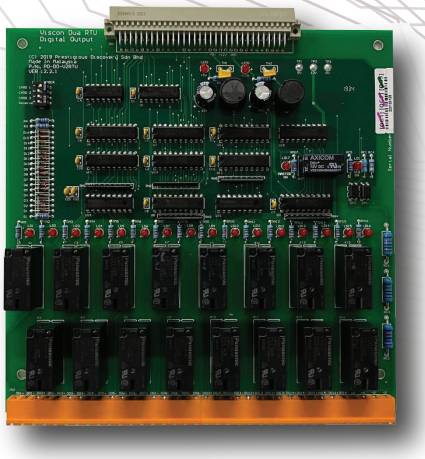
- Temperature: 0-70°C
- Relative Humidity: ≤ 95%

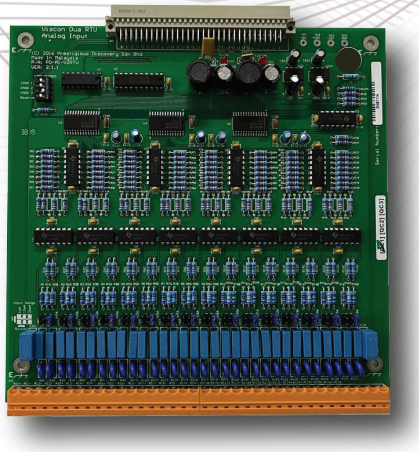
Other features

- Real-time clock (RTC)
- Watchdog timer

Technical Brief: RTU Input/Output Cards







Digital Input Card

- Configurable as wetting or dry contact
- Each DI point functions independently of any other points and can be assigned as any type of digital point.
- Each DI point is optically isolated

Input Channels

- Per card
  - Digital inputs: Up to 32 points
- Per module
  - Digital inputs: Up to 96 points
- Card per module
  - Digital inputs: Up to 3 cards

Input Signal Level:

- Nominal voltage: 24 VDC
  - Level Range: +10... +34
  - V- Input Current: typically
  - 2mA/input
- Nominal voltage: 48 VDC
  - Level range: +38... +58
  - V- Input Current: typically
  - 2mA/input
- Nominal voltage: 110 VDC
  - Level range: +100..120V
  - Input current: typically
  - 2mA/input

Mechanical Layout

- Printed circuit board: 186x190mm
- Subrack: 2U x 19″

Other Features

- Status/alarm with priority
- Sequence of event (SOE) with time tag for up to 256 events
- Binary coded decimal
- Pulse accumulator – form C and form A
- Single and double bit
- Contact debounce handler
- Relay chattering handler
- Digital input invert capability
- Noise immunity to 50 Hz at 3VAC

Digital Output Card

- Configurable as wetting or dry contact
- Programmable pulse duration
- Configurable control command time-out period
- Each DO point is optically isolated

Input Channels

- Per card
  - Outputs: 16 outputs for dry contact
  - 16 outputs for wetting contact
- Per module
  - Outputs: 48 outputs for dry contact
  - 48 outputs for wetting contact
- Card per module
  - Outputs: Up to 3 cards

Potential isolation :

- Yes, group-wise, from one another and against power supply.

Relay Characteristics

- Contact Rating:
  - 150W/1250VA
  - 250VAC, 30VDC
- Isolation Rating:
  - 2kV (Open contact)
  - 10kV (Contact - Coil)
  - 6kV (Contact - Contact)

Output Type

- Trip/Close
- Raise/Lower
- Programmable pulse resolution (100 msec by default)

Modes of Operation

- Execute immediate
- Arm before execute

Analogue Input Card

- Sixteen analog measured values can be connected to the AI card.
- Configurable measuring range
- Dual pole connection
- Differential inputs are protected against static and dynamic over–voltage by a protection circuit
- Designed to handle the continuous current looping during logic module replacement
- Each AI point is optically isolate

Input Channels

- Per card
  - Inputs : up to 16 differential inputs
- Per module
  - Inputs : up to 48 differential inputs (3 cards x 16 inputs)
- Measuring Ranges :
  - + 1 mA
  - + 10 mA
  - 0-20 mA, 4-20mA
  - + 5VDC
  - 0-10VDC
- Resolution :
  - 12bits unipolar (11 bit + 1 sign bipolar)
- Nonlinearity error
  - 1 LSB
- Quantization error
  - 1 LSB
- Gain Temp. Coefficient :
  - Unipolar 3 ppm/°C
  - Bipolar 5 ppm/°C
- Accuracy = <0.1% full scale at 25°C
  - Common Mode Rejection Ratio = 90dB
  - Normal (differential) Mode Rejection Ratio = 60 dB

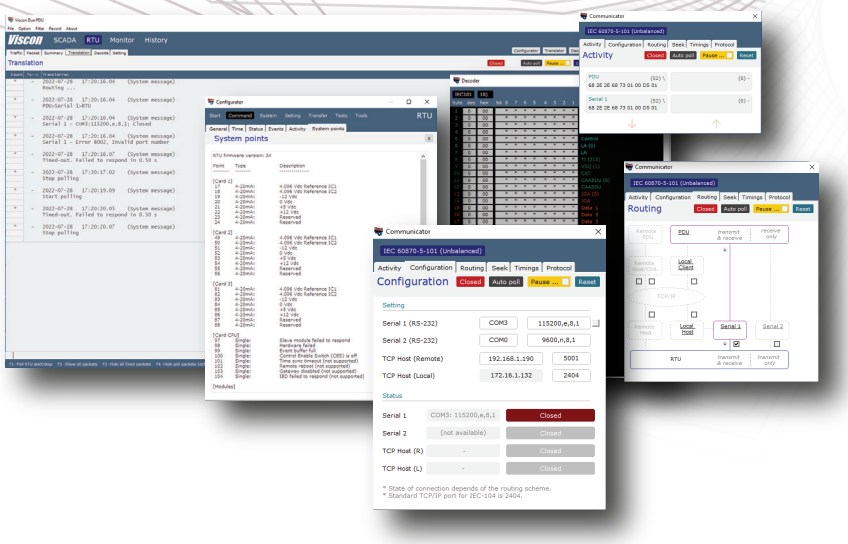
Optional features :

- Average
- Over-range
- Minimum
- Maximum
- Rate of change

Programming & Diagnostic Utility (PDU)

The Programming and Diagnostic Utility (PDU) is a Windows compatible software which enables user to issue all commands to the RTU. Its primary function is to setup up the RTU's configuration and simulating commands from SCADA Master.

It is connected to COM2 (Diagnostic port) of the RTU. It is a user friendly software that supports various standard Microsoft Windows features and facilities.



Standards

Environmental compatibility

- IEC 68-2-2
- IEC 68-2-30
- IEC 60255-5
- IEC 60255-21-1
- IEC 60255-21-2

Electromagnetic compatibility (EMC)

- IEC-1000-4-3
- DD ENV 50204
- CISPR 22 Class A (EN55022 Class A)
- IEC 1000-4-4
- IEC 1000-4-5
- IEC 1000-4-6
- IEC 255-5
- IEC 1000-4-2
- IEC 1000-4-8
- IEC 1000-4-10
- IEC 255-22-1
- IEC 60255-5

System Component

Standard items:

- RTU I/O Module
- CPU card
- Digital Input card
- Digital Output card
- Analogue Input card

Optional Item:

- HX-RTC Module
- Interlink Module (1U)
- PDM Interlink Module
- GPS Module

Application Examples

- Electrical distribution (SCADA) management system
- Platform/Plant automation system
- Building automation system
- Security systems, etc.

Communication Protocol

Our products support numerous data communication protocol and are capable to work in a multi-protocol environment. Listed are some of well-known globally accepted data communication protocols that are supported by our products.

Standard (Native) communication protocol

- IEC 60870-5-101
- IEC 60870-5-104

Other protocol implemented/supported

- DNP 3.0
- Modbus + Wave
- IEC 61131
- Others (Client Specified)

Ordering Infomation

Part number/Version	No. Description
PD-CPU-V2RTU/Ver 2.0	Central Processing Unit card
PD-DI-V2RTU/Ver 2.0	Digital Input card
PD-DO-V2RTU/Ver 2.0	Digital Output card
PD-AI-V2RTU/Ver 2.0	Analogue Input card
PD-PDU-V2RTU/Ver 2.0	Programming and Diagnostic Utility Software v2.xx

PRESTIGIOUS DISCOVERY SDN BHD (286874-W)
10, Jalan Sri Rejang 5, Rampai Business Park South,
53300 Setapak, Kuala Lumpur, Malaysia.
Tel: +603 4149 7006 Fax: +603 4149 6945

www.viscon.com.my
info@viscon.com.my
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